

ARULAI AH KAJANTHAN

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SUMMARY

Software Engineering undergraduate with strong interests in Embedded Systems, IoT, Machine Learning, and Full-Stack Development. Experienced in designing and developing real-time IoT applications, AI-powered intelligent systems, and scalable web platforms using technologies including ESP32, Raspberry Pi, MQTT, React, Node.js, Python, Firebase, and Machine Learning frameworks. Skilled in sensor integration, wireless communication, cloud-connected systems, and AI-based solutions, with a passion for building innovative technologies that solve real-world problems.

EDUCATION

University Of Sri Jayewardenepura
B.Comp Honours in Software Engineering

Colombo, Sri Lanka
2023 - Present

J/Velanai Central College
Ordinary Levels & Advanced Levels

Jaffna, Sri Lanka
Completed

PROJECTS

Smart Fingerprint Attendance System Using ESP32 | *ESP32, Node.js, React, Firebase, MQTT* | 🐙🔗

- Developed an IoT-based biometric attendance system using ESP32 and AS608 fingerprint sensor
- Implemented offline attendance storage using EEPROM with automatic synchronization after reconnection
- Created module-wise and timetable-based attendance analytics with a real-time reporting dashboard using React
- Built system workflow: ESP32 → HiveMQ Cloud (MQTT) → Node.js Server → Firebase → React Dashboard
- Designed a real-time attendance synchronization architecture using MQTT protocol and Firebase Realtime Database

CowGuard – Smart Livestock Monitoring System | *ESP32, LoRa SX1278, GPS, MQTT, Firebase, Node.js, React* | 🐙🔗

- Developed an IoT-based livestock monitoring system using ESP32, GPS, and LoRa for real-time cattle tracking.
- Designed the data pipeline: LoRa Tracker → ESP32 Base Station → MQTT → Node.js → Firebase → React Dashboard.
- Implemented live GPS tracking, geofencing, device monitoring, and email-based alert notifications.
- Built a Web dashboard for real-time livestock tracking, device management, and cloud-based monitoring.

Smart Accident Detection and Real-Time Alert System (FYP - Ongoing) | *ML, OpenCV, GPS, IoT*

- Developing a smart digital meter prototype for motorcycles with real-time navigation, accident detection, and emergency alert capabilities.
- Implementing a hybrid accident detection system using sensor fusion (IMU, vibration, GPS, heartbeat) and computer vision for improved accuracy.
- Designing a cloud-based emergency notification system with real-time GPS tracking, rider verification, and community-based alerts.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, JavaScript

Embedded Systems & IoT: ESP32, MQTT, LoRa, UART, SPI, I2C, Sensor Integration

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, YOLO, Computer Vision, RAG, LangChain

Frontend Development: React.js, Next.js, HTML, CSS, Tailwind CSS

Backend Development: Node.js, Express.js, REST APIs

Databases & Cloud: Firebase Realtime Database, MongoDB

Communication Protocols: MQTT, HTTP, UART, SPI, I2C, LoRa Communication

Developer Tools: Git, GitHub, VS Code, PlatformIO, Postman

Operating Systems: Linux, Windows

REFERENCES

Dr. S. Majuran

Senior Lecturer, Department of Knowledge Engineering and Communication

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University of Sri Jayewardenepura